Lloyds Metals and Energy Limited

Expansion of Iron Ore Beneficiation,Sponge Iron, Captive Power Plant and Installation of New Unit of Iron Ore Crushing and Screening Plant, Iron Pellets Plant and Coal Gasifier

at

MIDC Konsari, Village: Konsari, Tahsil: Chamorshi, Dist: Gadchiroli, Maharashtra

TECHNICAL SPECIFICATION FOR MOTORISED GATE

Doc. No: 2021-022-A00-ME-0001

Rev-0

1.00.00 INTENT OF SPECIFICATION

This Tender Specification is intended to cover the design, engineering, manufacture, supply, pre-assembly / assembly, tests at manufacturer's works, shop painting, forwarding (duly packed for transportation), transportation, delivery to the site, complete services of erection, testing at site, successful commissioning, operation & training of Owner's personnel, trial run, final painting, freight and transit insurance, all taxes, duties, octroi and other charges as applicable and handing over the system in a flawless operating condition to purchaser, for all equipment, drives & accessories for 'Motorised Sliding Gate' as specified herein and elsewhere in this Bid Document and as required for safe, trouble free, smooth and efficient operation of the gate which is to be installed at the boundary of the proposed Iron Ore Beneficiation Plant of Lloyds Metals and Energy Limited (LMEL), at MIDC Konsari, Village Konsari, Tahsil Chamorshi, District Gadchiroli, Maharashtra.

2.00.00 GENERAL INFORMATION TO TENDERER

2.01.00 **Lloyds Metals and Energy Limited (LMEL)** has proposed the expansion of their Iron ore beneficiation plant, Sponge Iron, Power plant and installation of New Unit of Iron ore crushing and screening plant, Iron Ore pellet plant and Coal Gasifier at MIDC Konsari, Village Konsari, Tahsil Chamorshi, District Gadchiroli, Maharashtra.

2.02.00 **Thriveni Earth Mover Private Limited (TEMPL)** in association with Lloyds Metals and Energy Limited (LMEL) would take up the expansion activity in the premises of an existing plant located at MIDC Konsari.

2.03.00 Around 126 acres of land has been identified for the above expansion project. The entire plant would be enclosed by a 2.4 m high boundary wall. Three motorised gates have been proposed in the east of the plant boundary to provide access from NH353A. Out of three gates, two gates would be utilised for the material movement of the plant whereas the third gate which would be placed between the two would be used as Main Entrance for the plant personnels. Refer to the layout as enclosed.

2.04.00 **Plant Location & Site Approach:**

Nearest Highway: Aashti Road at 50 m in the East direction.
Nearest Airport: Nagpur, 166 Km: in the North West
Nearest Railway Station: Kelzar ,34 Km : in the North West
Nearest Village: Konsari 0.5 Km : SSW

A. Technical Datasheet for Double Leaf Automatic Swing Gates

Datasheet-1

Technical Specification	Description
Type	Double Leaf Automatic Swing Gates
Duty Class	Intensive Use
Working cycle	80%
Location	Main Entrance
Qty	1 No.
Gate Dimension	
Clear Width in mm.	10450 mm
Height in mm	2400 mm
No. of Leaves	Two (2)
Gate Structure	
Outer Frame	Mild Steel
Internal members	Mild Steel
Hinges	High precision top & bottom pivoted hinges with ball bearing / heavy duty hinges
Material of Construction	
Outer Frame	Mild Steel
Internal Members	Mild Steel
Hinge	Suitable for intensive operation.
Details for Automation System	
Drive Unit	Motor with Actuator arm-CE certified /Under Ground Gear Operated
Power Supply	230 V
Travel control system	Mechanical stop
Opening time	Adjustable/90 Deg
Maximum opening	90 Deg
Accessories	

Flashing Light	For Visual Signaling during Opening & Closing
Station Push Button	Open / Emergency - Stop / Close controls. Plastic housing IP 65
Safety Sensor	The Gate remains open as long as the sensor works
Salient Features	
	Equipment shall be CE marked according to the European Machinery Directive and Standard European Norm
	Manual operation in case of power failure/ motor break down.
	Motor shall be protected by an internal thermal switch.
	Electronic control unit permits/allows operation time adjustment, partial opening, automatic closing,
	Step by step control, stop safety, reverse safety.
	Should be easily operable by security persons.
	Shall be suitable for operating in a dusty environment.

B. Technical Datasheet for Single Leaf Motorised Sliding Gates

Datasheet-2

Technical Specification	**Description**
Type	Single Leaf Motorised Sliding Gates
Duty Class	Intensive Use
Working cycle	80%
Location	Material Entry & Exit Gate
Qty	2 Nos.
Gate Dimension	
Clear Width in mm.	10450 mm
Height in mm	2400 mm
No. of Leaves	One (1)
Gate Structure	

Outer Frame	Heavy Duty tubular box section
Internal members	Heavy Duty tubular box section
Guide Rollers	Guide rollers shall be mounted on both side of column providing lateral support & smooth movement to the gate leaf
Material of Construction	
Outer Frame	Mild Steel
Internal Members	Mild Steel
Bottom Rail	C channel & round polished bar
Bottom Rollers	Wheels with ball bearings and axle with lubrication point
Details for Automation System	
Drive Unit	Motor with Rack & pinion
Travel control system	Limit Switch
Opening time	10-15 Sec
Maximum opening	Full aperture of the Gate
Accessories	
Flashing Light	For Visual Signaling during Opening & Closing
Station Push Button	Open / Emergency - Stop / Close controls. Plastic housing IP 65
Safety Sensor	The Gate remains open as long as the sensor works
Salient Features	
	Equipment shall be CE marked according to the European Machinery Directive and Standard European Norm
	Manual operation in case of power failure/motor failure.
	Motor shall be protected by an internal thermal switch.
	Electronic control unit permits/allows operation time adjustment, partial opening, automatic closing,
	Step by step control, stop safety, reverse safety.
	Should be easily operable by security persons.
	Shall be suitable for operating in a dusty environment.